Evolutionary Intelligence (2026) 19:90
https://doi.org/10.1007/s12065-026-01205-3

RESEARCH PAPER

Quantum hidden Markov model for sequential analysis

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Received: 20 August 2024 / Revised: 19 December 2025 / Accepted: 2 May 2026
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Abstract
Machine learning focuses on making computer systems learn from data and make decisions based on acquired knowledge. Hidden Markov model (HMM) is a probabilistic model in machine learning for representing and processing sequential data. To enhance the performance of sequential analysis, we propose a quantum hidden Markov model (QHMM) that transforms each component of classical HMM into its quantum counterparts. The proposed system consists of an encoding method that converts emission probabilities of observation onto quantum states and utilizes Gram-Schmidt orthonormalization to convert transition matrices into quantum operators. Moreover, it employs mid circuit measurement to carry hidden-state information across time steps. To evaluate the performance of the proposed QHMM, we apply part-of-speech (POS) tagging on English corpora and the English-Hindi code-mixed dataset. A comparison between the performance of the proposed model and the classical HMM has also been shown, where QHMM provides higher F1-score compared to classical HMM.

Keywords Quantum computing · Machine learning · Hidden Markov model · Part of speech tagging · Quantum hidden Markov model · Natural language processing

1 Introduction

Machine learning [1] is a branch of artificial intelligence [2] that allows computer systems to learn patterns from data without human intervention. It has achieved significant success in various domains such as image processing, bioinformatics [3, 4], and natural language processing (NLP) [5]. However, the performance of machine learning models is highly dependent on large amounts of data and complex algorithms. Processing such vast data requires substantial resources and extensive computation, which is difficult for classical computing. Many real-world problems, particularly those involving sequential and stochastic dynamics, demand high-dimensional modeling.

On the other hand, quantum computing [6] is a promising computing paradigm that follows the principle of quantum theory to process information efficiently. It has the potential to solve complex problems that are difficult for classical computing. For specific problems, quantum computing performs significantly faster and more efficiently than classical methods. For example, Shor's algorithm [7] for the prime factorization of a large number shows exponential speedup compared to classical algorithms. This hope has led to an increase in interest in researching the possibilities of integrating the principles of quantum physics with classical machine learning algorithms to address the limitations of classical ML. This combination of quantum computing and machine learning is known as quantum machine learning [8, 9], which focuses on utilizing the benefits of quantum mechanics' properties, leading to enhanced machine learning algorithms.

Among sequential learning models, the Hidden Markov Model (HMM) [10] has been a cornerstone in modeling probabilistic temporal dependencies. The fundamental aspect of HMM is the use of probabilistic transition between hidden states and the generation of observable events from

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Published online: 28 May 2026

those hidden states. The transition between hidden states within HMM enables them to handle uncertainties and variations inherent in real-world data. Consequently, HMM has become a powerful tool for various tasks such as recognition of speech and handwriting [11], bioinformatics [3, 4] and part-of-speech tagging [12]. Despite its utility, the HMM presents challenges, including the scalability issue, as the number of parameters grows quadratically with the number of states, making both inference and training computationally expensive [13].

In particular, QML has the potential to address the challenges of HMM. In this context, Monras et al. [14] introduce a hidden quantum Markov model (QHMM. and show that it is more expressive than HMM. Moreover, Srinivasan et al. [15] proposed QHMM learning algorithms that demonstrate superior performance compared to Baum-Welch HMM learning algorithms. However, there is still a lack of a QHMM framework to perform sequential analysis on a real data set because they proposed theoretically that they show the theoretical advantage of QHMM. However, a practical QHMM framework for performing sequential analysis on realworld datasets is still lacking. Moreover, the existing proposed QHMM model focuses primarily on the theoretical formulation and their potential advantages, without demonstrating empirical validation or real-data implementation.

To overcome this gap, we propose a QHMM model that converts classical HMM components into their quantum counterparts. Our main contributions are: (1) we design a novel QHMM framework that employs data encoding methods to convert emission vectors as the quantum state and transition matrix into a quantum operation by utilizing Gram-Schmidt orthonormalization; (2) we employ partial measurement to obtain outcome and the ancilla register acts as a persistent quantum memory that carries hidden-state information across time steps. (3) Finally, the QHMM model is applied to perform POS tagging on two distinct datasets: code-mixed English-Hindi data and English corpora. In addition, this study presents a comparative analysis of the performance of the proposed QHMM and its classical counterpart for POS tagging.

The structure of the paper is as follows: Sect. 2 explains the essential details related to HMM and quantum computing. Section 3 explains the comprehensive review of the literature on QHMM demonstrated by several authors. Our proposed system is elucidated in Sect. 4, where we present the conversion of classical HMM components into quantum frameworks. The description of the database is given in Sect. 5, followed by the use of POS tagging on both sets of data, and the results achieved and a comparative analysis in Sect. 6. Finally, the paper is concluded in Sect. 7 with the final observations.

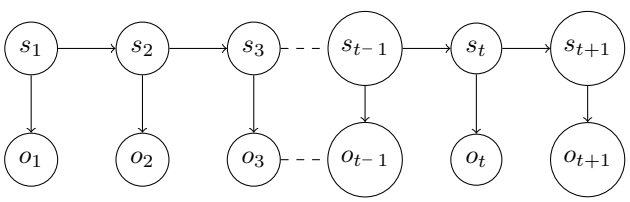


Fig. 1 Hidden Markov model

2 Background

2.1 Hidden Markov model

HMM is a probabilistic approach that relies on a joint distribution of two stochastic processes to efficiently analyze and manage time-series data. The working of HMM is centered around the Markov property, which states that the future state depends only on the current state instead of the sequence of states appearing before it. HMMs consist of a set of hidden states along with observable events. The hidden states are not directly visible; the model makes inferences based on the observable events. Figure 1 shows the structure of HMMs in general.

The components of an HMM can be described as follows:

- Hidden States - A finite set of states that cannot be observed directly, represented as s_1, $s_2 ... s_{t+1}$.
- Observables - Sequence of visible events generated by the corresponding states, represented as o_1, $o_2 ... o_{t+1}$.
- Transition Probabilities - The probability that the system transitions from state i to state j. It is calculated as $P(s_{t+1} = j | s_t = i)$ where s_t denotes the current state of the system and s_{t+1} denotes the future state in which the system will transition.
- Emission Probabilities - The conditional probability of observing a value o from a state s. It is calculated as $P(o_t | s_t)$ where o_t represents the observation at time 't' and s_t denotes the hidden state.

The emission probabilities are typically organized into a matrix known as the emission probability matrix. Each element in the emission probability matrix E, represented as e_{ij}, gives the probability that j^{th} observation being emitted from the state *i* and is given as Equ. 1.

$$e_{ij} = P(o_t = j | s_t = i) \tag{1}$$

Like emission probabilities, transition probabilities are also organized in the transition matrix T. Each element of the matrix T represents the probability of moving from state *i* to state *j* and is defined as Equ. 2.

$$t_{ij} = P(s_{t+1} = j | s_t = i) \quad (2)$$

The transition and emission probability matrices are both stochastic matrices. In context with the transition probability matrix, the stochastic property ensures that the probabilities of transitioning from a particular state to all other states sum up to 1. In context with the emission probability matrix, the stochastic property ensures that the probabilities of observing plausible observations from a particular state sum up to 1.

2.2 Quantum states and principles

Quantum computing leverages the principles of quantum mechanics, such as superposition and entanglement, to process information efficiently. Unlike the classical world, which uses bits for representing information, quantum computers use quantum bits (qubits). The classical bit states for computation are 0 or 1, but a qubit can be 0, 1, or a combination of both. Traditionally, the possible state of qubits is denoted by the Dirac notation[16]: $|0\rangle$ and $|1\rangle$.

Superposition is a fundamental characteristic of quantum mechanics that allows a quantum system to coexist in multiple states simultaneously. For example, a qubit can exist in the possible states $|0\rangle$ and $|1\rangle$. Through superposition, the quantum state of a qubit can be represented as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are the complex amplitudes of the states $|0\rangle$ and $|1\rangle$, respectively, and their squared sum equals one. So, qubits can exist in multiple states simultaneously, which leads to quantum computers processing information faster than classical computers.

Entanglement is another quantum phenomenon that broadens the disparity between classical and quantum computations. It allows particles to be connected, and one particle's state influences another's state despite the distance between them. If a group of particles is entangled, then the quantum state of each particle is dependent on the others, i.e., their quantum states cannot be described independently. This property plays a vital role in quantum computing algorithms.

Quantum measurement is a fundamental concept for observing the possible value of qubits. It collapses a quantum system in a superposition of qubits into one possible outcome based on the probability associated with the individual particles. For example $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, it will collapse into state $|0\rangle$ with a probability of $|\alpha|^2$ and $|1\rangle$ with a probability of $|\beta|^2$, depending on the underlying calculations. Once a quantum system is measured, the classical output is post-processed based on the requirements, finally moving to evaluation.

The building blocks of quantum computing are the quantum gates and circuits. Unlike classical gates, quantum gates operate on qubits and are supposed to be reversible. This characteristic of quantum gates ensures that the operation of a quantum gate can be undone, and it is possible to always trace back to the input given the output. The reversibility of operators arises from the necessary principle of quantum mechanics, where a quantum state evolves based on unitary transformations ($U^\dagger U = UU^\dagger = I$), which are reversible. The unitary transformations preserve the norm of quantum states and ensure the generation of valid states after each operation.

3 Literature survey

HMM, first introduced by Leonard E. Baum and Ted Petrie [10] in the late 1960 to establish an efficient probabilistic framework to model sequential data where observations are visible but underlying states are not directly observable. For the past two decades, HMM gained popularity and widely used in various fields due to their flexibility, especially for tasks such as speech [17] and handwriting recognition [3] and bioinformatics [4]. Specifically, HMM has been used to perform POS tagging [12, 18–22] due to their ability to consider word likelihoods and contextual transitions between tags to generate accurate predictions. A significant advancement of HMM-based tagger is made by Thorsten Brants [18], who introduces the Trigrams'n'Tag (TnT) model that utilizes smoothing techniques and unknown word management demonstrating good performance on both English and German Corpora. Cutting et. al. [19] proposed further advanced HMM tagger that demonstrates high tagging accuracy 96% through efficient use of the Viterbi and Baum-Welch Algorithm. These works establish HMM as an efficient framework to perform sequential analysis. However, their computational requirements grow rapidly with increasing vocabulary size and state complexity that motivate the requirement of a more scalable methodology.

Besides this, quantum computing has emerged as a powerful computing paradigm that exploits the principle of quantum mechanics and is able to handle the limitations of classical machine learning algorithms and enhance the performance of machine learning algorithms. Recently, several works demonstrate the potential of quantum computing across different machine learning algorithms. Such as Gong et al. [23] introduced a quantum convolutional neural network that utilized variational quantum circuits and achieved faster convergence and higher accuracy compared to classical convolutional neural networks on MNIST dataset. Similarly, Zhou et al. [24] introduced quantum generative adversarial networks that show superior performance than their classical counterpart. Chen et al. [25] provide the first proof principle that VQC can implement a quantum

reinforcement learning framework to approximate deep Q-learning. Beyond learning tasks, quantum computing can also improve the performance of optimization problems, such as the quantum particle swarm optimization algorithm based on diversity migration [26]. These advantages highlight the potential of quantum computing to accelerate machine learning tasks.

Taking into account these improvements, various researchers have investigated quantum computing principles for HMMs. Their works mainly focus on overcoming the limitations and enhancing the expressiveness of classical HMM. First, Monras et al. [27] introduced the hidden quantum Markov model, where hidden states are quantum states and transitions between those hidden states defined by Kraus operators. Moreover, they demonstrate that QHMM has more representational efficiency, since certain stochastic processes can be modeled using fewer hidden states compared to classical HMM. Clark et al. [28] later highlighted the connection between QHMM and open quantum systems with instantaneous feedback. The authors demonstrated that these instances align with the concept of QHMM, leveraging quantum feedback control mechanisms. The authors have shown that the output sequences observed from the open quantum systems can be used to model stochastic processes effectively and thus can be interpreted as QHMMs.

Subsequently, Srinivasan et al. [29] demonstrated how classical HMM can be simulated in quantum circuits and formulate the quantum counterpart of HMM. In addition, the authors introduced a learning algorithm for estimating parameters from the data within this quantum framework. Further, Adhikary et al. [30] demonstrated a theoretical foundation for QHMM, elaborating on their expressiveness and learning them from data. The authors have shown QHMM as a subclass of observable operator models that are special since they do not suffer from the negative probability problem owing to their design. Recently, Li et al. [31] introduced another approach to implement QHMM that utilizes the conditional master equation to show the interconnections among the internal states of the quantum system. Markov et al. [32] first, who formulated a quantum circuit ansatz and gave a training algorithm. The proposed test cases were evaluated by running the sequences on a quantum simulator and quantum hardware.

Despite these theoretical advantages, most existing QHMMs models focus mainly on theoretical formulations and conceptual frameworks, with little emphasis on practical implementation or validation using real-world data. Since then, there remains a noticeable gap between the theoretical development of QHMMs and their applicability to real sequential learning tasks. To address this gap, we introduce a QHMM framework capable of processing real datasets and performing end-to-end sequence modeling. To our best knowledge, this work is the first to apply a practically implemented QHMM to the task of POS tagging on real dataset.

4 Proposed model

This study presents a quantum machine learning algorithm, QHMM to perform sequential analysis. The proposed system has been divided into two modules: data encoding quantum operation. Fig. 2 represents systematic architecture of QHMM to serve as a decoding framework for sequential data analysis. In this implementation, each component of the classical HMM, such as emission probabilities of observations, and the transition matrix, is mapped to its quantum counterpart. The architecture of QHMM utilizes two sets of quantum registers: an ancillary register and a qubit register. It begins by encoding the emission vector of the first observations using a qubit register. Then, a quantum operator $GS(A)$ stand for Gram-Schmidt orthonormalization, is applied to the encoded quantum state. Finally, a partial measurement is made to measure only the qubit register, which

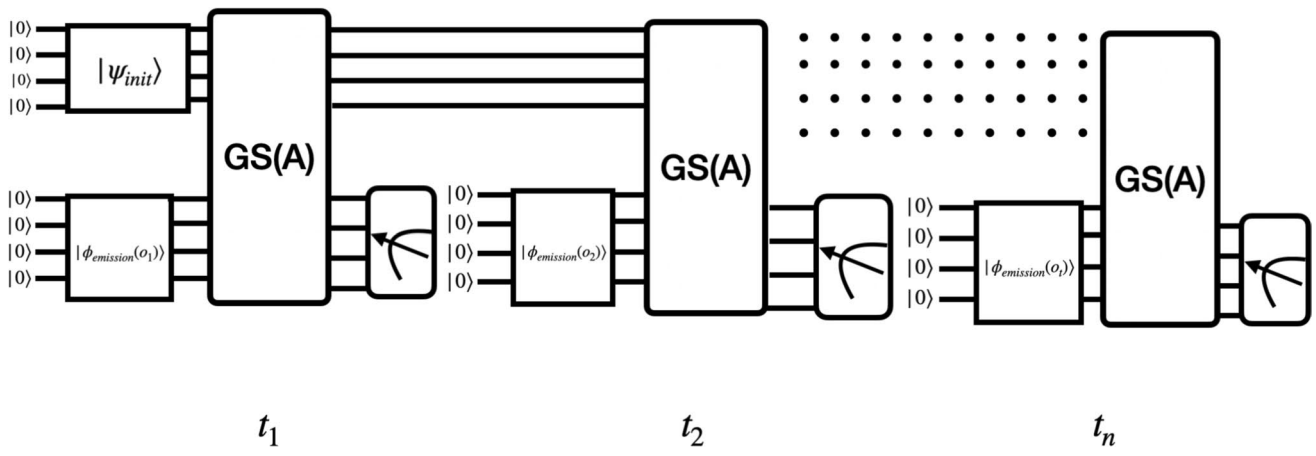


Fig. 2 The archiotecture of proposed QHMM

produces the observed output. Afterward, the emission qubits are reset and encoded with the emission vector for the following observation. Meanwhile, the ancillary register remains unchanged, retaining and propagating the quantum information of the previous hidden state to the subsequent step.

The detailed explanation of the QHMM framework is provided in subsequent subsections such as data encoding in subsection 4.1, quantum operation 4.2, and quantum circuit 4.3. It starts with data encoding, which gives a detailed explanation of how the emission probability of observations is encoded in quantum data. In the next subsection, quantum operations explain how the transition matrix is transformed into a unitary matrix. Finally, the partial measurement outlines how measurements are performed and encodes the emission probability of the next observations.

4.1 Data encoding

Working with classical data is not possible when dealing with quantum models. We must encode the data to process it and make it compatible with the particular quantum model. The process by which classical data are represented and efficiently put through a quantum system is known as data encoding. Several methods have been discussed theoretically in recent years; two commonly discussed methods by researchers are angle and amplitude encoding. In this work, we have considered both these methods for encoding the classical probabilities into a quantum state space and chosen amplitude encoding based on compatibility with our proposed model.

The amplitude encoding [33] represents classical data such that each data point can be represented as the amplitudes of a quantum state. To encode input data of size N, amplitude encoding needs a quantum state of n bits, where $(n \geq log_2 N)$. For example, if X represents the data point of size N that will be encoded in a quantum state, then the required quantum state $|\psi_x\rangle$ can be defined as shown in Equ. 3. α is the normalization constant; x_i denotes the elements of the original data vector that contain emission probabilities; N is the number of elements in the vector; and $|i\rangle$ is the computational basis states.

$$|\psi\rangle = \alpha \sum_{i=1}^{N} x_i |i\rangle \tag{3}$$

We have an emission vector for each observation taken from the emission probability matrix. Each emission vector is then normalized and encoded in the amplitudes of a quantum state created by a combination of qubits depending on the dimension of the emission vectors. The encoded quantum state will have its amplitudes corresponding to the normalized data points.

Input: Emission Probability vector for an observation.
Output: Quantum State representing the observation.
procedure DATA ENCODING($EmissionProbabilities$)
    $sum \leftarrow 0$
    for each probability p in $EmissionProbabilities$ **do**
        $sum \leftarrow sum + \sqrt{p}$
    end for
    $NormalizationConstant \leftarrow sum$
    for each probability p in $EmissionProbabilities$ **do**
        $p \leftarrow \sqrt{p}/NormalizationConstant$
    end for
    $Qubits \leftarrow \lceil \log_2(\text{length of } probabilities) \rceil$
    Prepare the required Quantum State.
    Integrate the state in the Quantum Circuit.
    return The prepared quantum state.
end procedure

Algorithm 1 Data encoding of emission probabilities of observations

Another encoding method, Angle encoding [34] uses quantum rotational gates to represent classical data in quantum states. It encodes one data point at a time, unlike amplitude encoding, which can represent multiple data points. If a data set has N features, angle encoding will require at least N qubits since each feature is encoded into the rotation angle of a single qubit.

However, Current quantum hardware is still in the early stages of development. Thus, to facilitate working with limited qubit resources, quantum circuits are designed to minimize the number of required qubits. To reduce the width of

quantum circuits, minimizing the number of parallel operations is necessary, which will also reduce the chances of facing errors and unnecessary quantum noise. Angle encoding is particularly efficient when working with data that are assigned to the range of rotation angles. Still, working with multiple data points requires more qubits for angle encoding than amplitude encoding. This study utilizes amplitude encoding in the QHMM framework instead of angle encoding for this particular reason.

Now, quantum data that is encoded by data encoding becomes the input of the quantum operations module explained in the next subsection, where the encoded observation is evolved according to the quantum counterpart of the classical transition matrix.

4.2 Quantum operation

The encoding step converts the emission probability of each observation into quantum states and serves as input to the quantum operation stage of the QHMM. In quantum machine learning algorithms, the process of encoding classical data is followed by quantum operations. Quantum operations are reversible and are represented by unitary matrices that always transform a quantum state into another valid one by preserving the norm of quantum states, ensuring no information is lost during the computation.

In the classical HMM framework, a transition matrix governs the evolution of hidden states. To achieve the changes according to the classical framework utilizing quantum operation, it is necessary to convert the classical operators into unitary matrices. So, we need to transform the transition matrix into a quantum operation and ensure that the unitary property holds. The proposed model uses Gram-Schmidt orthonormalization [35] to convert a transition matrix into a unitary matrix. As shown in algorithm 2, the method ensures that each column vector of the newly created matrix is orthogonal to others and has a magnitude of 1 by eliminating the respective projections. Ensuring orthonormality is essential because the resulting unitary matrix will represent a valid evolution of the quantum state, where each state is distinct and the probabilities will sum up to 1.

Input: Transition Probability Matrix T
Output: Unitary Matrix U
function GRAM-SCHMIDT(T)
    Initialize $\hat{U}$ as an empty matrix
    for `each column` t_i `in` T **do**
        Copy t_i to $\hat{u}_i$ in $\hat{U}$
        for `each` $\hat{u}_j$ `before` $\hat{u}_i$ `in` $\hat{U}$ **do**
            Orthogonalize $\hat{u}_i$ against $\hat{u}_j$
            Normalize $\hat{u}_i$
        end for
    end for
    return $\hat{U}$
end function
function VERIFY UNITARY PROPERTY($\hat{U}$)
    Compute U by normalizing each column of $\hat{U}$
    Check if $UU^{\dagger}$ results in an identity matrix
    return Boolean if U is unitary
end function
function QUANTUM OPERATION(U)
    Use U as a quantum gate in quantum circuit
end function

Algorithm 2 Converting a matrix into unitary matrix

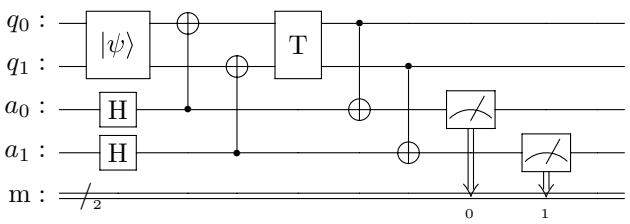


Fig. 3 The proposed circuit for QHMM

The transformed unitary operator is applied to the complete quantum circuit architecture, which is presented in the next subsection. It interacts with the joint quantum state of the encoded quantum state and auxiliary qubits to produce sequential dynamics of the QHMM.

Table 1 Universal POS tagset

Tag	Meaning
ADJ	Adjective
ADP	Adposition
ADV	Adverb
CONJ	Conjunction
DET	Determiner, article
NOUN	Noun
NUM	Numeral
PRT	Particle
PRON	Pronoun
VERB	Verb
P	Punctuation marks
X	Other

Table 2 Code-mixed tagset

Category	Type	Description
Noun (G_N)	N_NN	Common noun
	N_NNV	Verbal noun
	N_NST	Spatio-temporal
	N_NNP	Proper noun
Pronoun (G_PRP)	PR_PRP	Personal
	PR_PRL	Relative
	PR_PRF	Reflexive
	PR_PRC	Reciprocal
	PR_PRQ	Wh-Word
Verb (G_V)	V_VM	Main
	V_VAUX	Auxiliary
Adjective (G_J)	JJ	Adjective
Adverb (G_R)	RB_ALC	Locative adverb
	RB_AMN	Adverb of manner
Demonstrative (G_PRP)	DM_DMD	Absolute
	DM_DMI	Indefinite
	DM_DMQ	Wh-word
Quantifier (G_SYM)	QT_QTF	General
	QT_QTC	Cardinal
	QT_QTO	Ordinal
Particles (G_PRT)	RP_RPD	Default
	RP_NEG	Negation
	RP_INTF	Intensifier
	RP_INJ	Interjection
Residual (G_X)	RD_RDF	Foreign word
	RD_SYM	Symbol
	RD_PUNC	Punctuation
	RD_UNK	Unknown
Conjunction	CC	Conjunction
Numeral	$	Numeral
Determiner	DT	Determiner
Pre- & postposition	PSP	Pre-/postposition
Others (G_X)	@	At-mention
	~	Re-tweet/discourse
	E	Emoticon
	U	URL or email
	#	Hashtag

4.3 Quantum circuit

Figure 3 presents a quantum circuit designed for the proposed system. It describes how encoded quantum states and the unitary transition operator are integrated and perform the state evolution through partial measurement. It consists of several components: qubits, ancilla qubits, classical registers, data encoding, quantum operation, measurement, and fundamental quantum gates. The number of qubits is decided by the size of the data needed to encode and the method of encoding used. In this work, we use amplitude encoding, which requires $log_2 N$ qubits to encode N data points.

Ancilla qubits are not part of the main qubits but are essential for quantum circuits because of their ability to facilitate partial measurements. In the quantum world, if a state is measured, it collapses into one of the possible states, thus resulting in a classical outcome and losing the state of the entire system. To tackle this problem, when both preserving the state of the system and measurement are necessary, ancilla qubits can be introduced to enable partial measurement, which will benefit the cause. Additionally, ancilla qubits are also used to assist operations like error correction, building complex gates, etc. In our case, we use auxiliary qubits to carry previous information throughout the circuit.

Classical registers are used to store the classical outcome derived after measuring the state of a system to make the result available for post-processing or to find out the result of the quantum operation. Quantum operations are used in circuits to transform the state of the system. A custom quantum gate has been designed from the transition probability matrix, facilitating the state transitions in the QHMM and effectively formulating the Markov process. In circuit 1, the quantum gate T enables the state transitions in the system.

The proposed circuit operates sequentially to process one observation at a time. For each time instance, the emission vector of an observation is encoded into quantum state and then the main qubit and the ancilla qubits are entangled using CNOT operations. This entanglement distributes probability amplitudes across possible hidden-state configurations, allowing the circuit to represent all valid transitions simultaneously. Next quantum operation, unitary transition gate T, which is transformed form transition matrix is applied to update the hidden-state representation. After this, inverse CNOT operations disentangle the ancilla qubits from the data qubits, ensuring that only the main qubits are measured. This enables partial measurement: the main qubits collapse to produce the observable output, while the ancilla qubits retain the hidden-state information required for the next iteration.

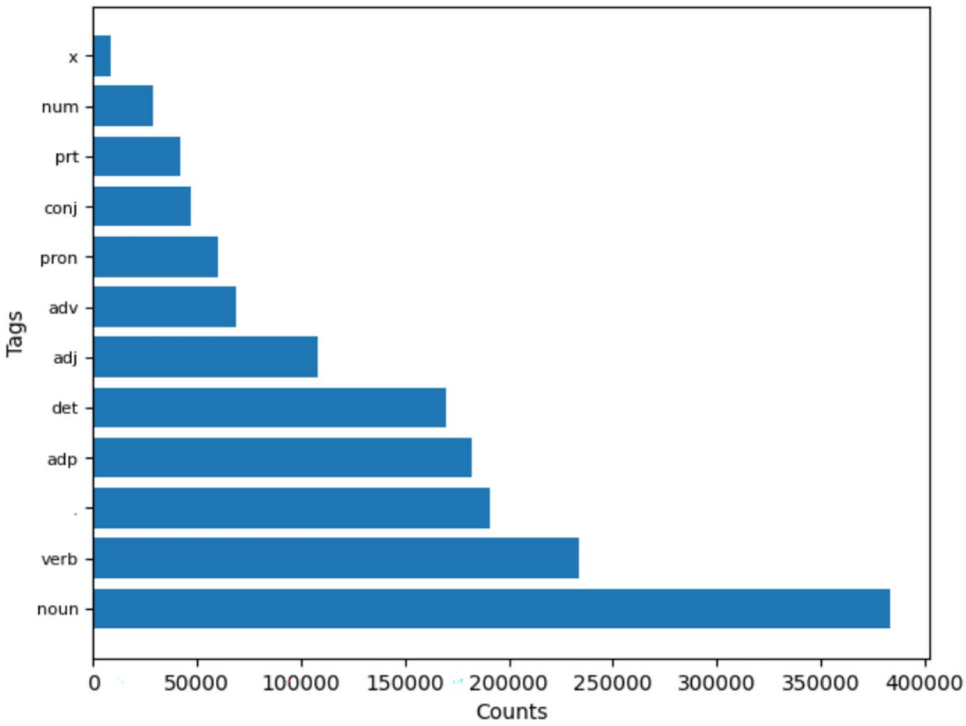


Fig. 4 Tag frequency of English corpora

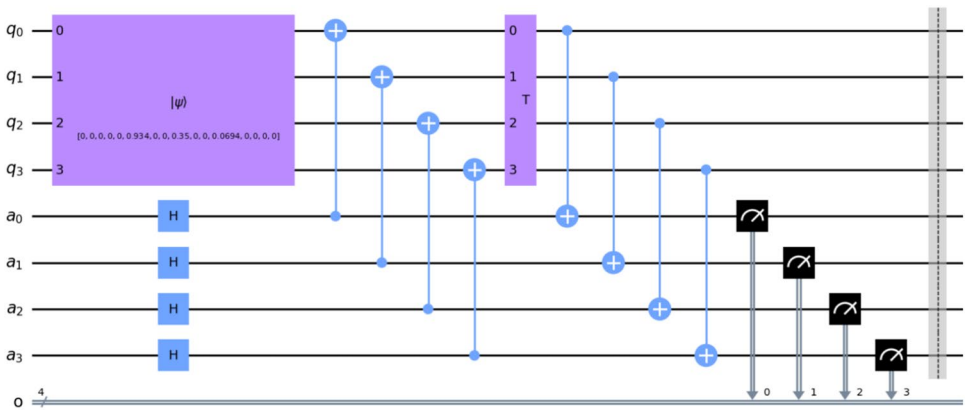


Fig. 5 Quantum circuit for POS tagging for English corpora

Table 3 Classification report of the evaluation of QHMM with the English Corpora

Tags	Precision	Recall	f1-score
P	0.96	0.98	0.97
ADJ	0.7	0.84	0.76
ADP	0.9	0.9	0.9
ADV	0.73	0.88	0.8
CONJ	0.85	0.98	0.91
DET	0.86	0.93	0.89
NOUN	0.96	0.83	0.89
NUM	0.89	0.84	0.86
PRON	0.89	0.71	0.79
PRT	0.74	0.96	0.83
VERB	0.91	0.9	0.9
X	0.03	0.5	0.05

4.4 Computational complexity of QHMM

The computational complexity of QHMM consists of two modules such as unitary construction and quantum state evolution during sequence processing. Converting the classical transition matrix $A \in \mathbb{R}^{N \times N}$ into a unitary matrix using the Gram–Schmidt process as $\mathcal{O}(N^3)$. Moreover, computing the evolution of the quantum state in each time demands $\mathcal{O}(T \log N)$., where T is the sequence length, and $\log N$ for each time instance encodes the observation. Therefore, overall complexity of the proposed model is $\mathcal{O}(T \log N + N^3)$

Table 4 Confusion matrix for the evaluation of QHMM with the English Corpora

	P	ADJ	ADP	ADV	CONJ	DET	NOUN	NUM	PRON	PRT	VERB	X
P	20627	0	2	121	47	4	0	207	3	20	2	16
ADJ	68	9520	152	568	102	75	414	85	37	98	191	44
ADP	48	338	16450	288	6	315	51	9	74	654	46	7
ADV	50	191	142	5066	79	20	21	18	4	162	6	6
CONJ	10	0	5	56	4270	0	1	28	0	0	0	1
DET	2	42	551	8	32	14410	96	0	55	343	7	2
NOUN	401	3136	622	535	326	623	44056	376	180	219	1822	389
NUM	195	130	161	194	135	81	74	6058	43	45	58	53
PRON	3	8	38	0	0	1201	26	6	3341	0	0	25
PRT	16	14	0	1	1	1	1	1	0	5481	114	0
VERB	44	208	238	67	31	45	1071	40	10	374	21732	35
X	1	1	0	7	0	0	2	0	0	6	0	17

5 Dataset description

We employ POS tagging on two distinct datasets to evaluate the suitability and adaptability of our proposed system.

5.1 English corpora

Natural language toolkit (NLTK) is a powerful library in Python that provides frameworks to work with linguistics and various NLP tasks such as text tokenization, parsing, stemming, and resources to work with natural languages. The NLTK data package [36] that has been used to train and test the baseline model consists of -

- 10% sample of the Penn Treebank [37] consisting of 3914 tagged sentences.
- 57340 tagged sentences from the Brown Corpus [38].
- 10948 tagged sentences from the Conll 2000 corpus [39].

The dataset has been split into train and test sets in a 9:1 ratio. The test set consists of 1,71,230 words to be tagged. The tagset has twelve unique tags, as shown in Table 1.

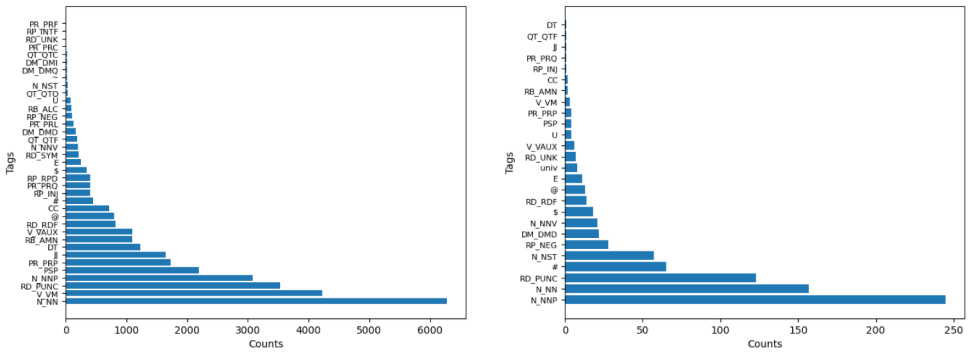
5.2 Code-mixed data

In multilingual societies, day-to-day conversations consist of a mix of several languages because people are usually fluent in several dialects, making code mixing [40] a very common phenomenon today. Thus, the corpus of collected social media data contains grammatical structures from several languages in sentences and phrases because speakers switch from one language to another within sentences. Due to the data that contain words from several languages, tagging the corpus is more complex.

In this study, we use social media code-mixed data [20] for evaluation, which consists of English-Hindi text in the corpus with code mixed below the level of the clause and both course- and fine-grained tag sets. We used the fine-grained tag set for evaluation. The details of the tags are shown in Table 2. The data consists of a total of 32,105 words in the train set and a total of 4116 words in the test set. Twelve universal tags are further divided to make the tag-set fine-grained, leading to 37 different tags.

6 Results and discussion

The model has been tested on the English Corpora and a code-mixed English-Hindi data set. This section contains detailed results from both experiments and a comparative



(a) Tag Frequency in Code-Mixed Data (b) POS Tag Distribution of Uncommon Words

Fig. 6 Insights on the distribution of the code-mixed data

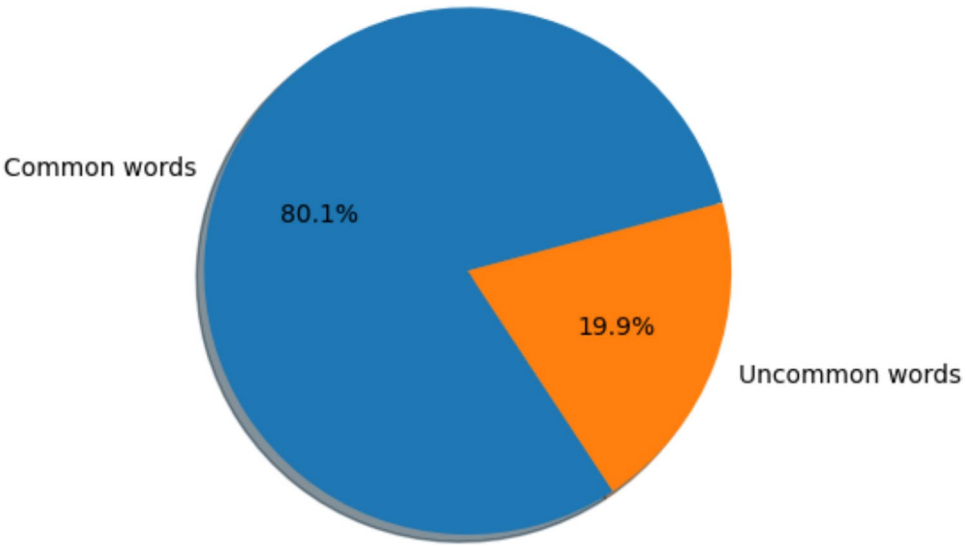


Fig. 7 Proportion of uncommon words in the test set

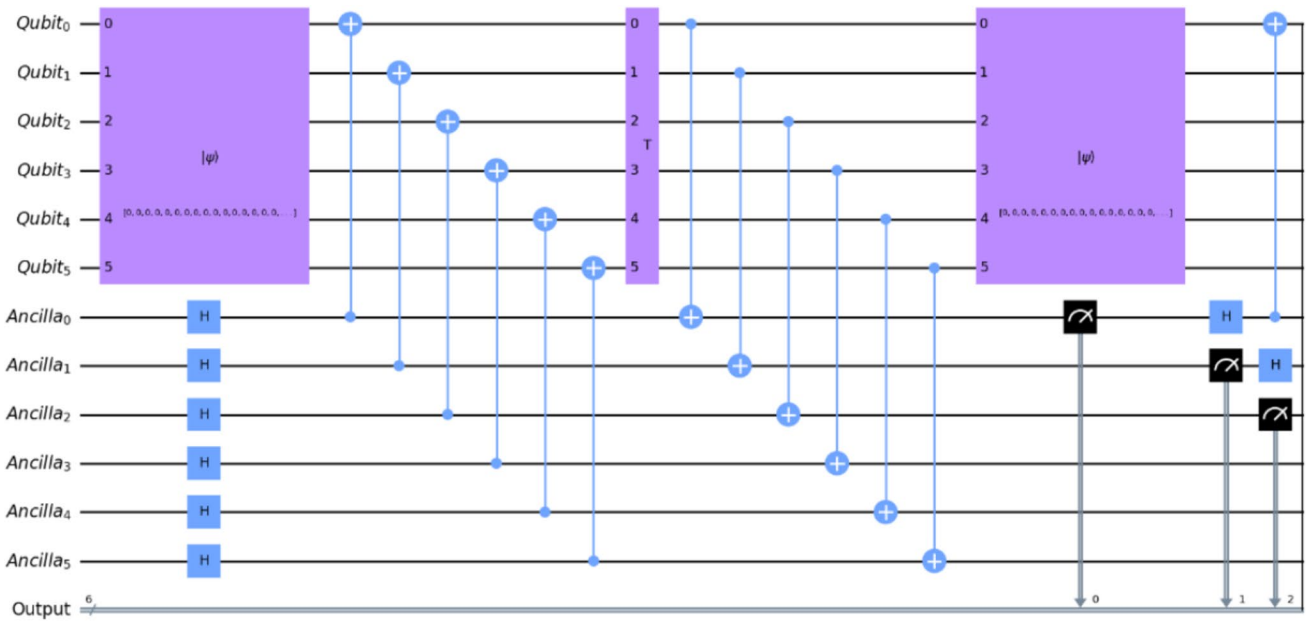


Fig. 8 A section of the QHMM circuit generated using Qiskit, for code-mixed data

Table 5 Classification report of QHMM on code-mixed data

Tags	Precision	Recall	f1-score
#	0.64	0.97	0.77
$	0.33	0.3	0.32
@	0.67	1.0	0.81
CC	0.44	0.67	0.53
DM_DMD	0.2	0.75	0.32
DM_DMI	0.06	1.0	0.11
DM_DMQ	0.03	1.0	0.06
DT	0.78	0.87	0.82
E	0.3	0.47	0.37
JJ	0.63	0.49	0.55
N_NN	0.83	0.45	0.58
N_NNP	0.66	0.4	0.49
N_NNV	0.06	0.12	0.08
N_NST	0.09	0.43	0.15
PR_PRC	0.17	1.0	0.3
PR_PRF	0.04	1.0	0.07
PR_PRL	0.26	0.82	0.4
PR_PRP	0.84	0.71	0.77
PR_PRQ	0.5	0.78	0.61
PSP	0.72	0.55	0.62
QT_QTC	0.1	0.75	0.18
QT_QTF	0.28	0.58	0.38
QT_QTO	0.12	0.83	0.22
RB_ALC	0.12	0.45	0.19
RB_AMN	0.64	0.56	0.6
RD_PUNC	0.78	0.9	0.83
RD_RDF	0.38	0.4	0.39
RD_SYM	0.11	0.09	0.1
RD_UNK	0.05	0.5	0.1
RP_INJ	0.26	0.55	0.35
RP_INTF	0.07	0.67	0.13
RP_NEG	0.18	0.6	0.28
RP_RPD	0.36	0.8	0.49
U	0.12	0.3	0.18
V_VAUX	0.42	0.61	0.5
V_VM	0.79	0.38	0.51
~	0.13	1.0	0.24 %

study between the performances of classical HMM and the proposed QHMM.

6.1 Evaluation parameters

To evaluate the performance of our experiments, we use various metrics, including accuracy, precision, F-1 score, recall, and detailed classification report. Additionally, we show the confusion matrix for both datasets. In the case of POS tagging, precision measures the number of times the model correctly assigned a particular tag, given the total number of times the model has predicted that tag during testing, and is calculated separately for each POS tag. Recall is used to understand the model's efficiency in identifying all occurrences of a particular POS tag. F1-Score checks the overall performance of the model and is a balance between both precision and recall. The confusion matrix is handy in assessing a model's performance regarding POS tagging since it consists of the number of times the model predicted a specific POS tag versus the number of times it occurs in the data.

6.2 Evaluation of QHMM with English corpora

The whole data is split on a 9:1 ratio for making up the train and test sets. A certain amount of imbalance might exist in the data collected from natural language since some tags are more frequent in speech and natural language than others. The English Corpora is also imbalanced owing to that particular reason, with instances of nouns being over 350,000 whereas adverbs occur less than 100,000 times, as shown in Fig. 4.

The quantum circuit for POS tagging of English corpora is shown in Fig. 5 and will be repeated n times, given the test set has a total of n words to be tagged. Given that the number of unique tags is twelve, amplitude encoding needs four main qubits for the effective data encoding and another set of four ancilla qubits. The T gate is the quantum gate facilitating the transition between states. The accuracy achieved by processing the data by the QHMM is 88.2 %. The classification report is detailed in Table 3.

The achieved accuracy of 88.2% indicates the effectiveness of QHMM in correctly identifying POS for words in the English-only text. Moreover, on a detailed breakdown of the confusion matrix provided in Table 4, it is clear that some parts-of-speech tags, such as Nouns, have high instances of getting identified correctly but have also been incorrectly tagged a certain amount of times as other POS. Meanwhile, the tag Particle has been precisely determined by the model mostly, except in a few instances where some were tagged as verbs instead. The detailed breakdown provided in Table 3, shows that the model has good precision and recall for POS like Verb, Conjunction, and Nouns, as demonstrated by the F1-scores which are close to 0.9 or above. Some POS have lower F1-scores, such as Adjectives and Pronouns, which indicate that the model needs improvement.

6.3 Code-mixed data exploration and visualization

The data being highly code-mixed, i.e., below phrase level posed challenges in the effective tagging of words by the QHMM. It also consists of a wide proportion of unknown words, which were never encountered in the train set. The frequency for each tag is shown in Fig. 6a. The model consists of 20% words uncommon to the train data as shown in Fig. 7, while Fig. 6b shows how frequently the model

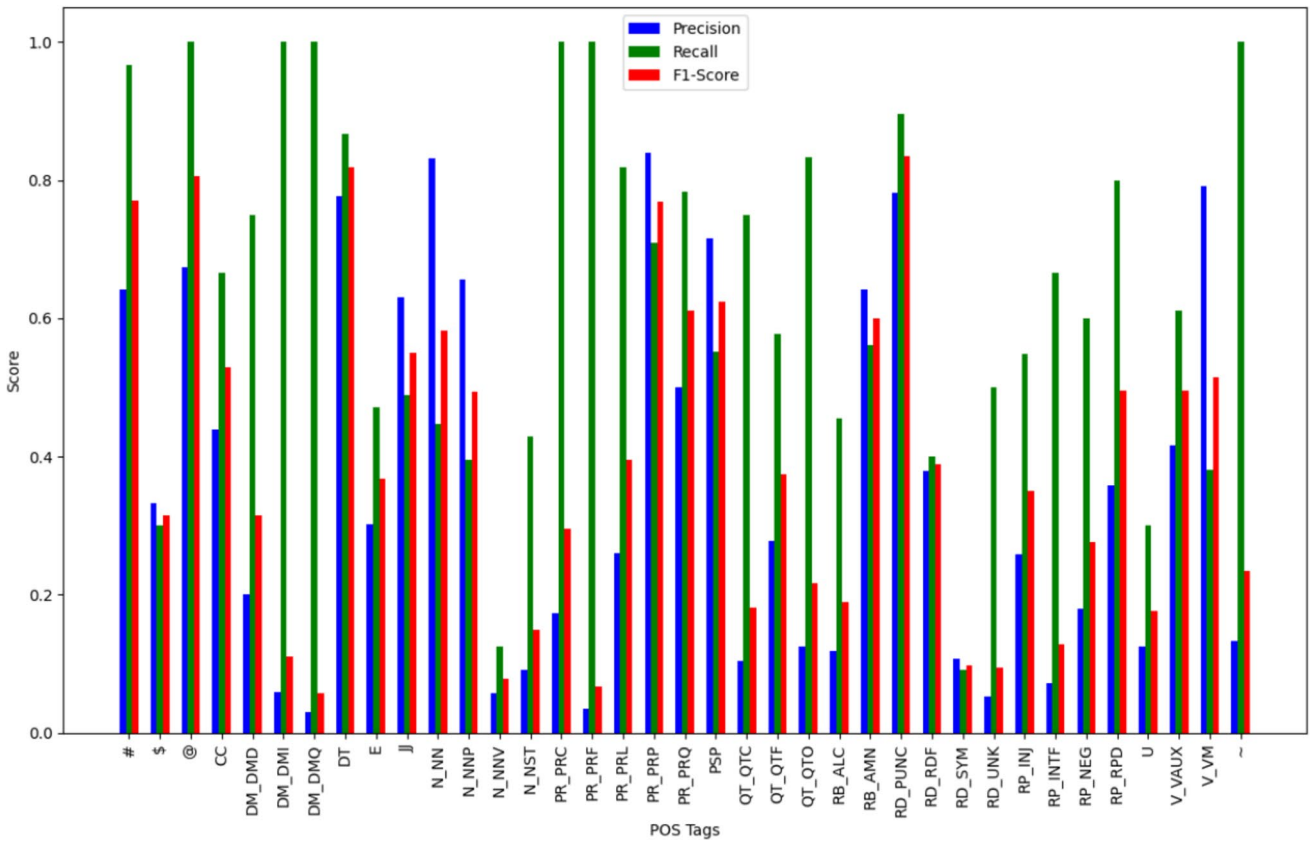


Fig. 9 Graphical representation of precision, recall and F1-scores for each POS tag in the Code-mixed data

encounters a word that is not present in the train set categorized by POS tag.

6.4 Evaluation of QHMM with code-mixed data

The code-mixed data has been divided into train and test sets in the ratio of 9:1 and the tag-set consisting of 37 unique tags has been used in the evaluation process. The test set, having a total of 4,116 untagged words has been processed using the QHMM. A portion of the quantum circuit used to process this data has been shown in Fig. 8. It goes through several repetitions depending on the length of the test set.

The accuracy achieved by QHMM for processing the code-mixed data is 57.2 %. As mentioned in 6.3, various issues were highlighted, which might have reduced this percentage compared to the evaluation of English corpora. One of the probable reasons behind this is the presence of multiple grammatical structures at the clause level. Detailed analysis of each specific POS tag is given by classification report in Table 5. The usage of the fine-grained taxonomy of tags for the evaluation of QHMM with code-mixed data has shown that the proposed model encounters difficulties when attempting to identify words as tags belonging to the division of the coarse-grained taxonomy of tags. However, it has been observed that the model performs significantly better when it assigns tags based on the coarse-grained tagset as demonstrated in the evaluation of QHMM using English corpora (Fig. 9).

6.5 Comparative analysis of HMM and QHMM

Classical HMM with Viterbi [41] and the proPOSed QHMM have been used to apply the POS tagging of code-mixed data to present a comparative analysis. Figure 10 compares the F1 scores for the respective POS tags for the classical HMM and the QHMM. The classical HMM achieves an accuracy of 58.96%. Compared on the basis of precision, it can be observed that in the case of several POS tags, the classical model performs better than QHMM, whereas QHMM achieves a better recall score in the case of some POS tags. Upon comparing the F1 scores of both models, in Fig. 10, it can be seen that QHMM has a higher score for many POS tags.

6.6 Observation

The proposed QHMM performs better or comparable performance in the data set compared to classical baseline HMM models. It gains higher confidence in frequently used tags and finds difficult for rare tags of words in code-mixed data set. In the case of monolingual data, QHMM effectively captures sequential dependencies and contextual relationships,

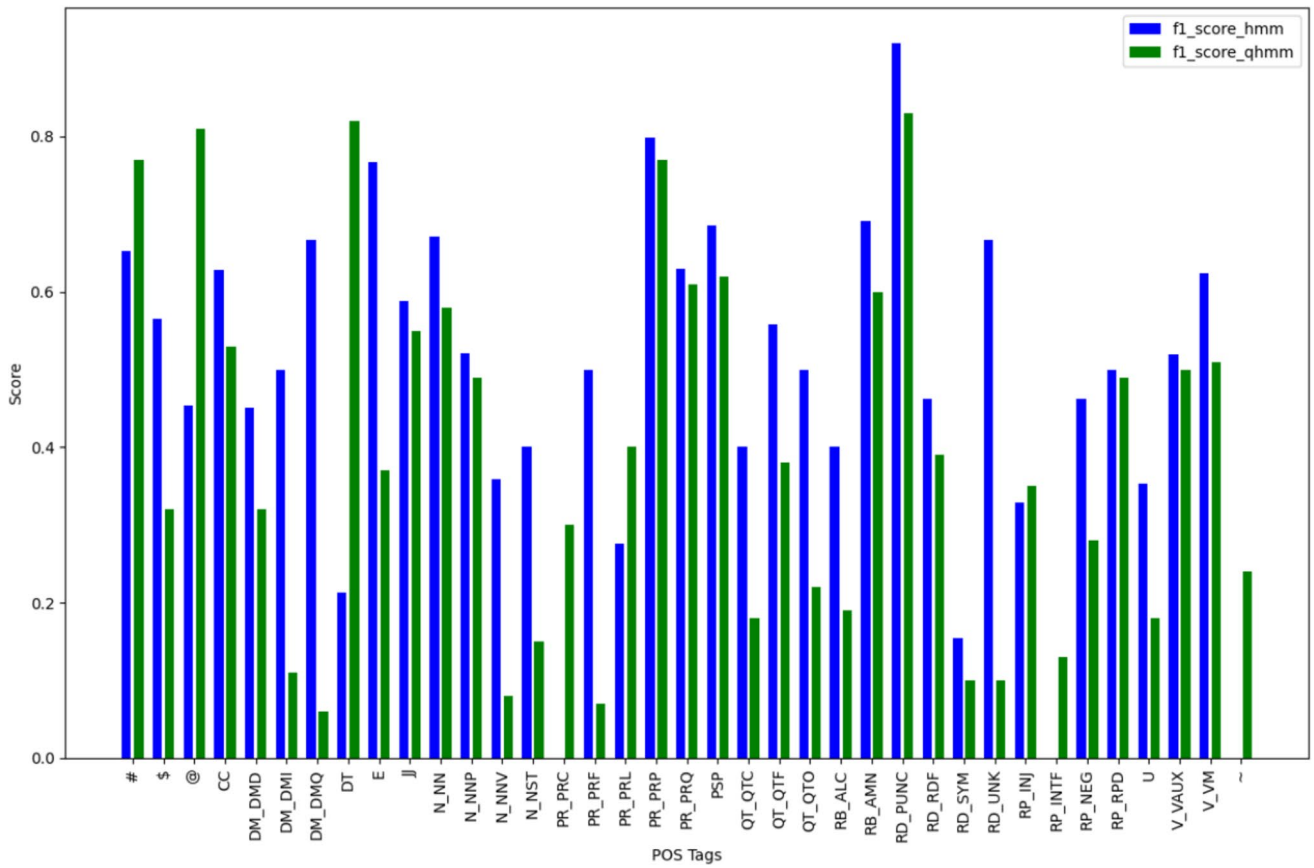


Fig. 10 Graphical representation comparing the F1 scores of HMM and QHMM on the code-mixed data

illustrating its strong capability for data-driven learning. Incorporating robustness on various datasets, the results highlight that QHMM offers a practical and generalizable framework for sequence labeling tasks.

7 Conclusion and future work

We proposed a novel QHMM model that employs amplitude encoding to encode the emission vector of the observation into the quantum state and utilize Gram-Schmidt orthonormalization to convert the transition matrix into a quantum operation. Moreover, the proposed employ partial measurement to get outcome at each time instance, and the ancilla register acts as persistent hidden states. To evaluate the performance of QHMM, two datasets, one monolingual English corpora and another mixed English-Hindi code data set, are applied for POS tagging. The evaluation of QHMMs, focusing mainly on code-mixed data, was carried out along with the comparison with the classical counterpart, which showed slightly higher accuracy. However, QHMM worked comparatively much better on the monolingual English data. In this paper, we utilize the proposed QHMM to perform decoding inference. Our future work will focus on performing the likelihood estimation and learning the parameters of QHMM. Moreover, we will apply the proposed QHMM to other sequential analysis such as Speech recognition.

Acknowledgements The work presented here falls under the Research Project Grant Ref. No. N-21/17/2020-NeGD supported by MeitY Quantum Computing Applications Lab (QCAL).

Author contributions Bitanuka Deb: Implementation of the system and paper writing. Shyambabu Pandey: Idea and paper writing. Partha Pakray: Overall Ideas, supervision and paper writing.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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